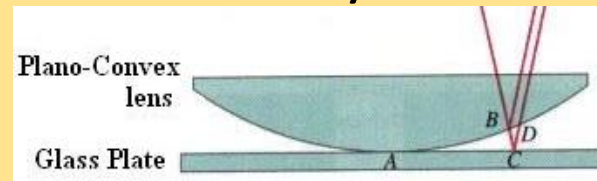


NEWTON'S RINGS-REFLECTED SYSTEM

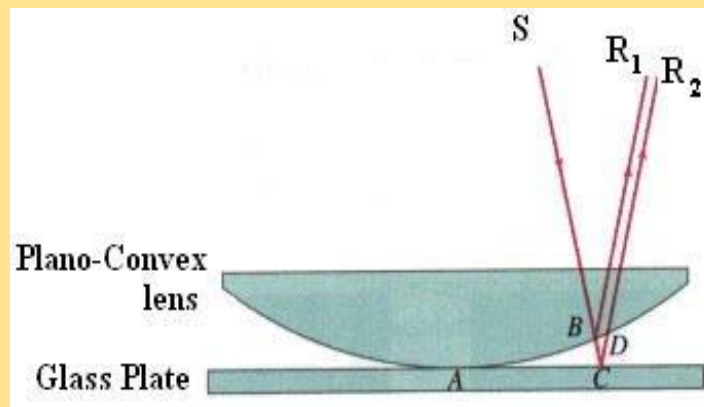
NEWTON'S RINGS

- Newton rings are example of fringes of equal thickness.
- They are observed when light is reflected from a plano-convex lens of a long focal length placed in contact with a plane glass plate.
- A thin air film is formed between the plate and the lens.
- The thickness of the air film varies from zero at the point of contact to some value t .
- If the lens plate system is illuminated with monochromatic light falling on it normally, concentric bright and dark interference rings are observed in reflected light.
- These circular fringes were discovered by Newton and are called Newton's rings.



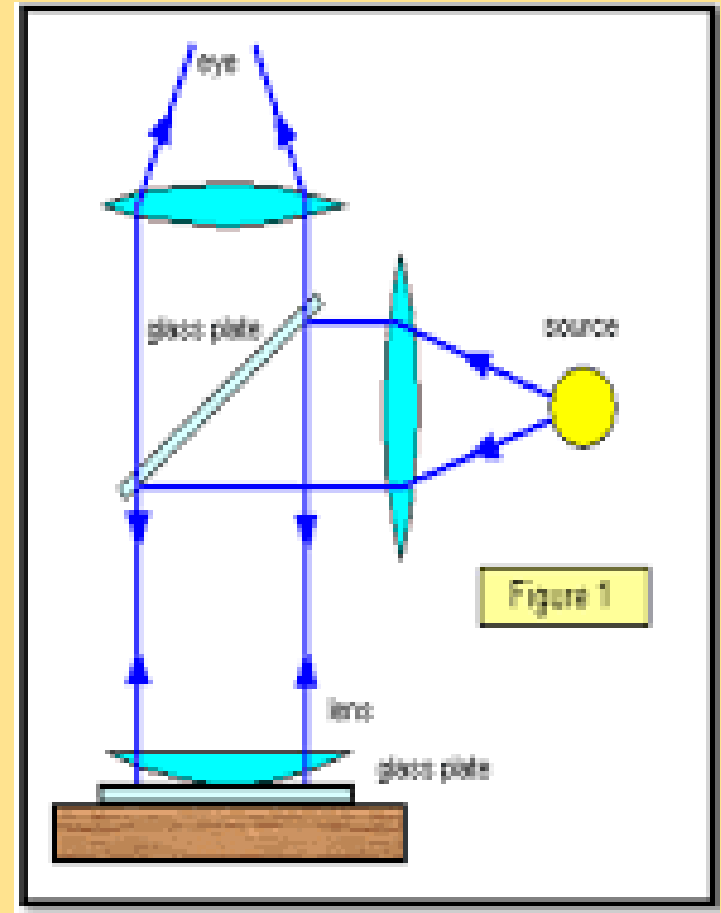
Experimental arrangement

- When a plano-convex lens with large radius of curvature is placed on a plane glass plate such that its curved surface faces the glass plate, a wedge air film (of gradually increasing thickness) is formed between the lens and the glass plate.
- The thickness of the air film is zero at the point of contact and gradually increases away from the point of contact.



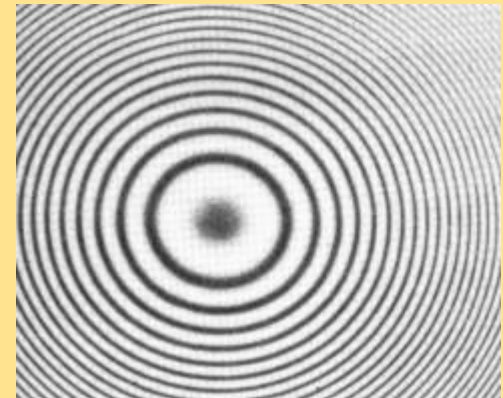
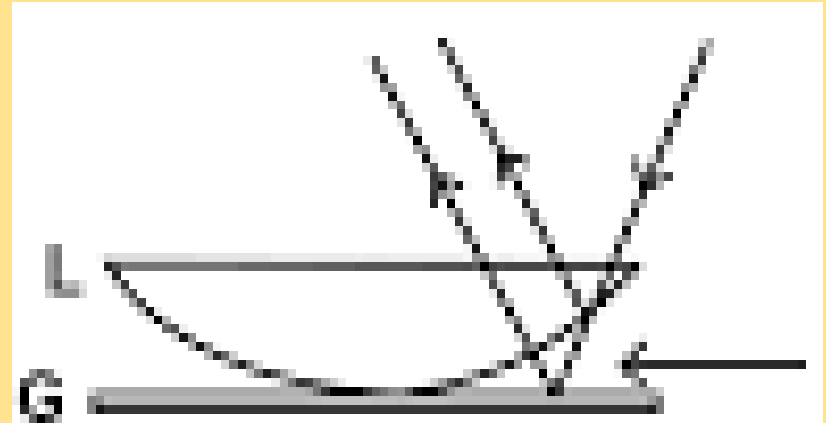
Experimental arrangement

- Monochromatic light from an extended source is rendered parallel by a lens
- It is incident on a glass plate inclined at 45 degree to the horizontal, and is reflected normally down on to a plano-convex lens placed on a flat glass plate
- Part of the light incident on the system is reflected from the glass to air boundary



Experimental arrangement

- The remainder of the light is transmitted through the air film
- It is again reflected from the air to glass boundary
- The two rays reflected from the top and bottom of the air film are derived through division of amplitude from the same incident ray and are coherent
- The two rays are close to each other and interfere to produce darkness or brightness



Concentric circular fringes are produced, uniform in thickness with the point of contact as the centre.

Formation of fringes

- When viewed with white light, the fringes are coloured.
- With monochromatic light, bright and dark circular fringes are produced.
- Here interference is due to the light reflected from the lower surface of the lens and upper surface of the glass plate.
- The fringes are circular because the air film is symmetrical (circular) about the point of contact of the lens with the plane glass plate
- The condition of brightness and darkness depends on the path difference between the two reflected light rays, which in turn depends on the thickness of the air film

Theory of Fringes:

We know that the rings are formed in reflected light (wedge shaped film) therefore the effective path difference between two interfering rays is

$$2\mu t \cos r + \frac{\lambda}{2} \dots\dots\dots (1)$$

- Where μ is the refractive index and t is the thickness of the film.
- In case of normal incidence and air film

$$\mu = 1, r = 0,$$

$$\cos r = 1$$

$$\text{Path difference} = 2\mu t + \frac{\lambda}{2} \dots\dots\dots (2)$$

$$\text{at a point of contact } t = 0$$

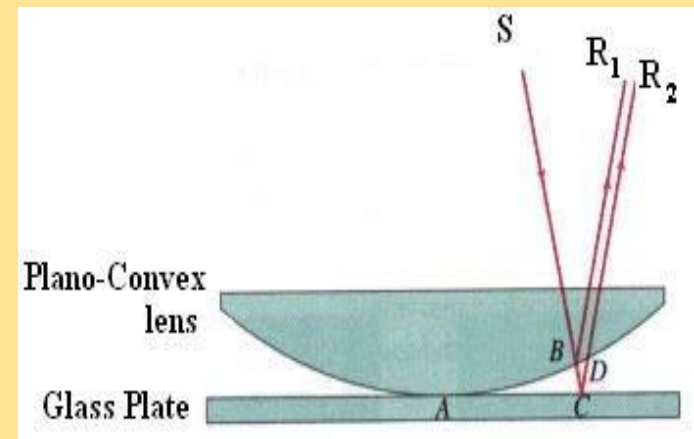
$$\text{Path difference} = \frac{\lambda}{2}$$

- This is the condition of minimum intensity hence center of the Newton ring is dark. But sometimes it is bright because dust particle come between the two surfaces at the point of contact.

Theory of Fringes:

The effective path difference between the two reflected rays R_1 and R_2 for a wedge shaped film from equation 2.18

$$\Delta = 2\mu t \cos(r + \theta) + \frac{\lambda}{2}$$



If the light is incident normally on the lens, $r = 0$ and near to point of contact θ is small; therefore near point of contact, $(r + \theta) \rightarrow 0^\circ$

Therefore
$$\Delta = 2\mu t + \frac{\lambda}{2} \dots 2.21$$

At point of contact $t = 0$ therefore the effective path difference $\Delta = \lambda/2$ which is odd multiple of $\lambda/2$. **Therefore the Central fringe is dark.**

Condition of Maxima (Bright Fringe): The effective path difference $\Delta = \pm n\lambda$; substituting this in equation 2.21

$$2\mu t + \frac{\lambda}{2} = \pm n\lambda$$

$$2\mu t = \pm \frac{(2n - 1)\lambda}{2} \dots 2.22$$

Condition for Minima (Dark Fringe): The effective path difference $\Delta = \pm \frac{(2n + 1)\lambda}{2}$; substituting this in equation 2.21

$$2\mu t + \frac{\lambda}{2} = \pm \frac{(2n + 1)\lambda}{2}$$

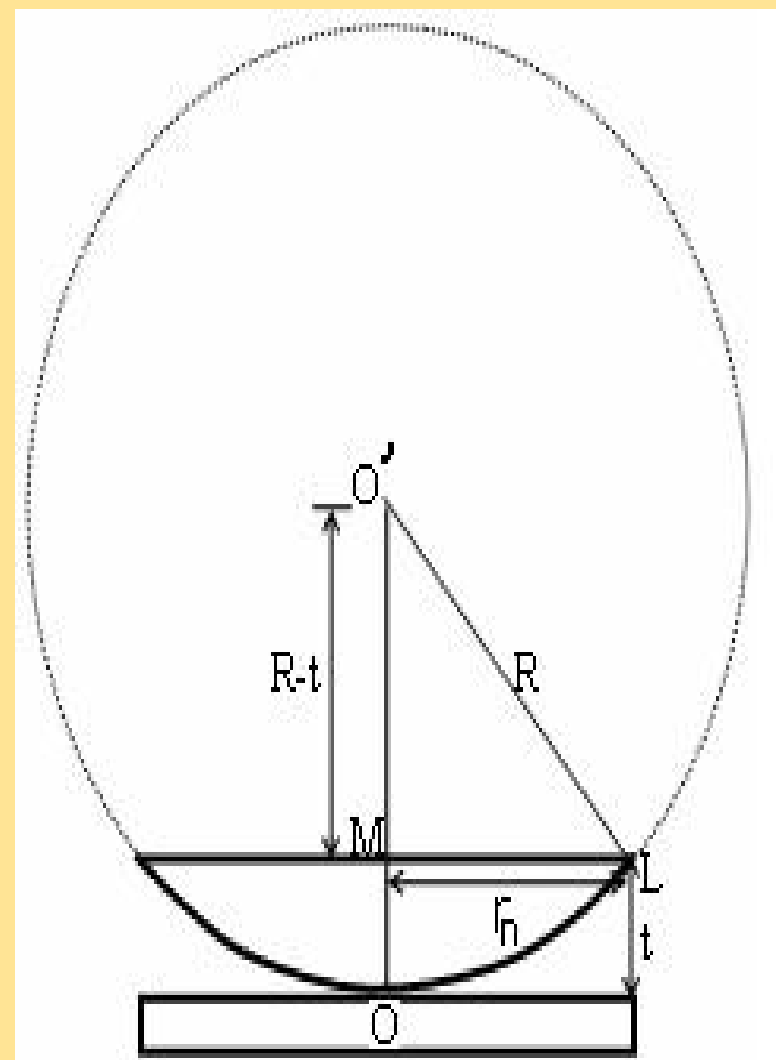
$$2\mu t = \pm n\lambda \dots 2.23$$

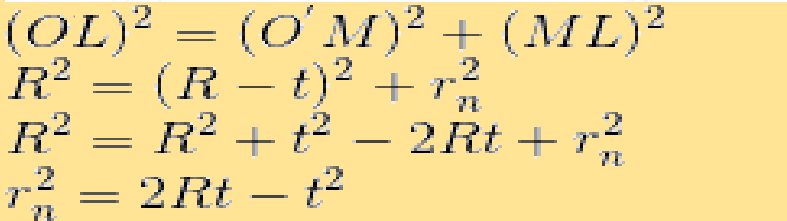
From 2.22 and 2.23 it is clear that for particular dark or bright fringe t should be constant.

Every fringe is the locus of points having equal thickness. Hence the fringes are circular in shape.

Diameter of Newton's Rings:

To calculate the diameter of fringes, assume a plano-convex lens is placed on a plane glass plate as shown in figure say R be the radius of curvature of lens. In $\triangle O'ML$





$$R^2 = (R - t)^2 + r_n^2$$

$$r_n^2 = 2Rt - t^2$$

$$r_g^2 = 2Rt$$

$$t = \frac{r_n^2}{2R} \quad \dots, 2.24$$

Diameter of Dark Rings

Substitute t from eq. 2.24 to eq. 2.23

$$2\mu \frac{r_n^2}{2R} = n\lambda$$

$$r_n^2 = n\lambda R / \mu$$

As $D_n = 2r_n$

$$D_n^2 = 4n\lambda R / \mu$$

$$D_n = \sqrt{4n\lambda R / \mu} \quad \dots 2.25a$$

The medium enclosed between the lens and glass plate is if air therefore, $\mu = 1$. The diameter of n^{th} order dark fringe will be

$$D_n = \sqrt{4n\lambda R} \quad \dots 2.25b$$

$$D_n \propto \sqrt{n}; \quad n = 0, 1, 2, 3, 4, \dots$$

The diameter of dark ring is proportional to square root of natural numbers

Diameter of Bright Rings

Substitute t from eq. 2.24 to eq. 2.22

$$2\mu \cdot \frac{r_n^2}{2R} = \frac{(2n-1)\lambda}{2}$$

$$r_n^2 = \frac{(2n-1)\lambda R}{2\mu}$$

As $D_n = 2r_n$

$$D_n^2 = 2(2n-1)\lambda R/\mu$$

$$D_n = \sqrt{2(2n-1)\lambda R/\mu} \quad \dots 2.26a$$

The medium enclosed between the lens and glass plate is if air therefore, $\mu = 1$. The diameter of n^{th} order bright fringe will be

$$D_n = \sqrt{2(2n-1)\lambda R} \quad \dots 2.26b.$$

$$D_n \propto \sqrt{(2n-1)}; n = 0, 1, 2, 3, 4, \dots$$

Spacing between Fringes

The Newton's rings are not equally spaced because the diameter of ring does not increase in the same proportion as the order of ring and rings get closer and closer as 'n' increases.

For example the diameter of dark ring is given by $D_n = \sqrt{4n\lambda R}$

Where $n = 0, 1, 2, 3, 4, \dots$

$$D_5 - D_4 = (\sqrt{5} - \sqrt{4})\sqrt{\lambda R} = 0.236\sqrt{\lambda R}$$

$$D_{25} - D_{24} = (\sqrt{25} - \sqrt{24})\sqrt{\lambda R} = 0.101\sqrt{\lambda R}$$

$$D_{15} - D_{14} = (\sqrt{15} - \sqrt{14})\sqrt{\lambda R} = 0.131\sqrt{\lambda R}$$

From above, we conclude that the fringe width reduces with increase in 'n'.

Newton's Ring with White Light

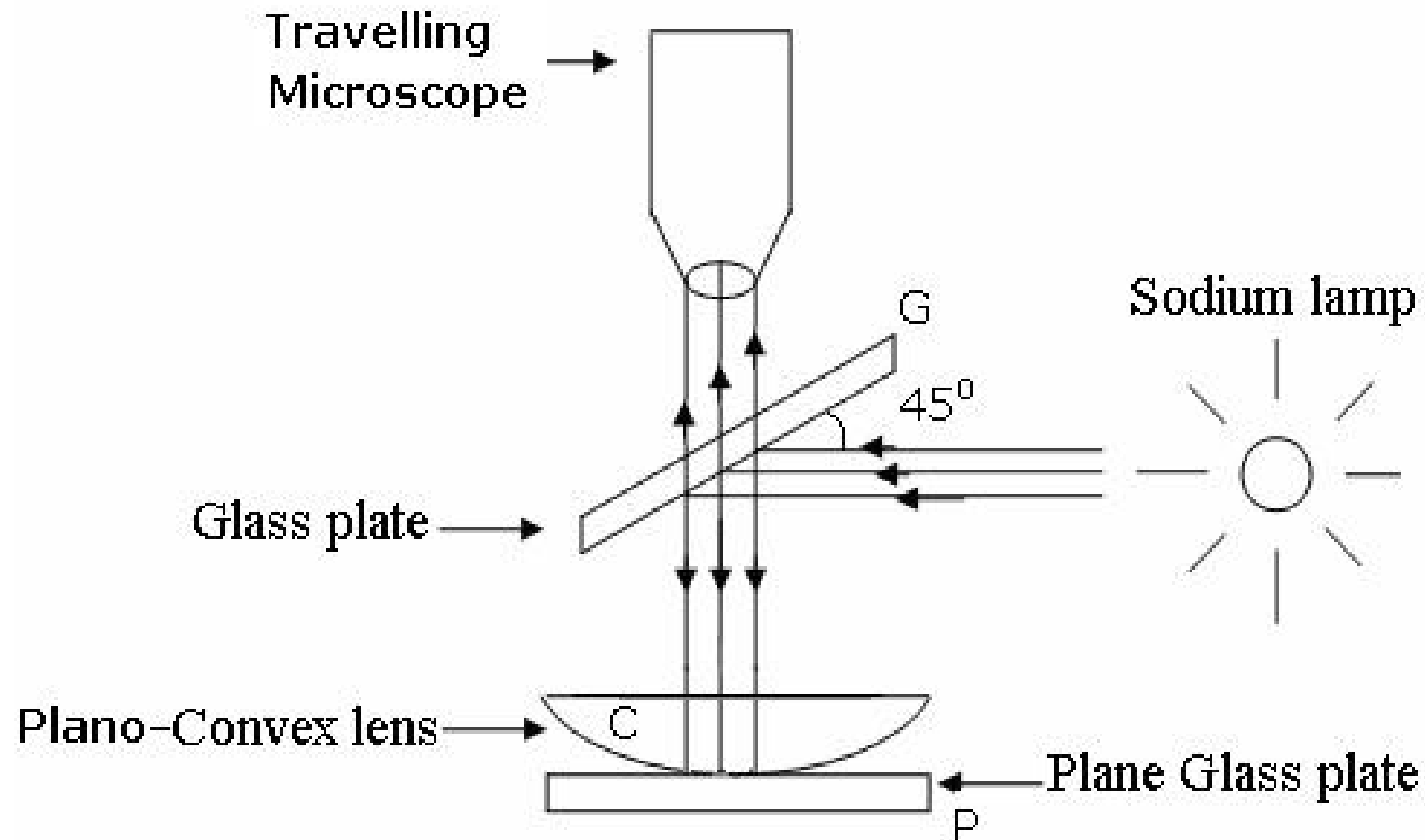
If the monochromatic source is replaced by the white light few coloured rings are seen around dark centre later illumination is seen in the field of view.



Applications of Newtons Ring

A. Determination of Wavelength of Light

In order to determine the wavelength of monochromatic source an experimental arrangement is set up as shown in figure.



The monochromatic light is made incident on the plano-convex lens and the reflected light is viewed in travelling microscope. Microscope is adjusted till the circular rings came in focus. Now the microscope crosswire is focused on the central dark spot and is moved slowly at one side, say right side. As the cross – wire moves in the field of view, dark rings are counted. The movement is stopped when it reaches at the 22nd dark ring. The cross wire is moved in opposite side and stopped at 20th dark ring such that the vertical cross-wire is made tangential to the ring. Thus starting from 20th the microscope position is noted down for 18th, 16th, 14th,6th ring. The microscope is quickly moved to the left side of the ring pattern and the positions of microscope is again noted down for 6th, 8th, 10th,20th ring. The diameter of the nth order ring is calculated by subtracting the left and right side position of the microscope. As we know that the square of diameter of nth dark ring is

$$D_n^2 = 4n\lambda R$$

Therefore the square of diameter of $(n+p)^{\text{th}}$ ring is

$$D_{n+p}^2 = 4(n+p)\lambda R$$

Subtracting both the above equation

$$D_{n+p}^2 - D_n^2 = 4(n+p)\lambda R - 4n\lambda R \dots (2.27)$$

Therefore

$$\lambda = \frac{D_{n+p}^2 - D_n^2}{4pR} \dots (2.28)$$

B. Determination of Refractive Index of liquid

In order to determine the refractive index of liquid the Newton's ring experiment is first performed for the air medium and the difference in the square of the diameter of $(n+p)^{\text{th}}$ and n^{th} dark ring is found as discussed above.

$$(D_{n+p}^2 - D_n^2)_{\text{air}} = 4p\lambda R \quad \dots 2.29$$

After this few drops of liquid of μ refractive index is placed on the glass plate. The plano-convex lens is then placed on the glass plate, as a result a film of liquid is formed between the lens and the plate.

The difference in the square of the diameter of $(n+p)^{\text{th}}$ and n^{th} dark ring is again calculated in the same manner for the liquid medium.

$$(D_{n+p}^2 - D_n^2)_{\text{liquid}} = \frac{4p\lambda R}{\mu} \quad \dots 2.30$$

Dividing equation 2.29 by 2.30, we get

$$\mu = \frac{(D_{n+p}^2 - D_n^2)_{\text{air}}}{(D_{n+p}^2 - D_n^2)_{\text{liquid}}} \quad \dots 2.31$$

Testing of optical planeness of surfaces

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A surface is said to be optically flat if it is plane upto $1/10$ th of the wavelength of light used.

- In order to test the planeness, we can use interference fringes obtained from an air film. The surface to be tested is placed in contact with an optically flat glass plate and the fringes are viewed.
- If the fringes of equal thickness are formed, then the surface is flat.
- Irregular and distorted fringe pattern is obtained if the surface is not flat.

Thank you